

- **Icons:** Lucide
- **State Management:** Zustand

Phase 1 — Product Discovery

You are acting as a Senior Software Architect, Senior Product Manager, Senior UI/UX Designer, and Senior Frontend Developer.

We are building a production-ready AI Placement Preparation Platform.

This project will be deployed on Vercel.

The platform should look and feel like a premium SaaS product rather than a college website.

Do NOT generate any code.

Your task is to create a complete Product Requirement Document (PRD).

Include:

- Product Vision
- Target Audience
- User Personas
- Problems Solved
- Market Analysis
- Feature List
- User Stories
- Functional Requirements
- Non Functional Requirements
- Success Metrics
- Monetization
- Future Roadmap
- Risks
- Technical Challenges

Everything should be production ready.

Output markdown only.

Phase 2 — Complete Architecture

Now design the complete software architecture.

The stack is:

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Next.js 15 App Router
React 19
TypeScript
Tailwind CSS v4
shadcn/ui
Supabase Authentication
Supabase PostgreSQL
Supabase Storage
Drizzle ORM
Vercel
Framer Motion
React Hook Form
Zod
Recharts

Design:

- Folder structure
- API architecture
- Route architecture
- Database schema
- Authentication flow
- Authorization
- Middleware
- Server Components
- Client Components
- Loading strategy
- Error handling
- Caching strategy
- File upload strategy
- Security
- Rate limiting
- Logging
- Environment variables

Do not generate code.

Only architecture.

Phase 3 — Database Design

Design a production-grade PostgreSQL database.

Tables should include:

Tables should include:

Users

Profiles

Resumes

Resume ATS Reports

Companies

Company Interview Questions

Company DSA Questions

DSA Problems

Daily Questions

Interview Questions

Bookmarks

Progress

Solved Questions

Notes

Placement Predictions

Activity Logs

Notifications

Sessions

Achievements

Create:

- ER Diagram (text)
- Relationships
- Indexes
- Constraints
- Cascade Rules
- Soft Delete Strategy
- Audit Fields
- Migration Strategy

Everything should support millions of users.

Phase 4 — Complete UI/UX

Design the entire UI.

The design language should be:

Modern

Minimal

Apple inspired

Glassmorphism

Smooth animations

Dark Mode first

Responsive

Accessible

Create complete page hierarchy.

Landing Page

Authentication

Dashboard

DSA

Companies

Interview Questions

Resume Analyzer

ATS Report

Profile

Settings

Bookmarks

Progress

Daily Challenge

Notifications

404

500

Loading

Skeleton Screens

Empty States

Dialogs

Sidebars

Navigation

Charts

Cards

Give complete component hierarchy.

No code.

Phase 5 — Landing Page

Design a world-class landing page.

Sections:

Hero

Problem

Solution

Section:

Features

Dashboard Preview

ATS Preview

Company Preparation

DSA Preparation

Interview Preparation

Testimonials

Pricing

FAQ

Footer

Animations

Scroll Effects

CTA

SEO Sections

Everything should be conversion optimized.

No code.

Phase 6 — Dashboard

Design a personalized dashboard.

Dashboard should show:

Welcome message

Current Target Company

Target Role

ATS Score

Resume Completion

Placement Probability

DSA Progress

Interview Progress

Solved Today

Current Streak

Daily DSA

Interview Question of the Day

Company Preparation

Weak Topics

Strong Topics

Graphs

Recent Activity

Recommended Questions

Recommended Interview Questions

AI Suggestions

Upcoming Goals

Notifications

Quick Actions

Recent Resume Analysis

Everything should adapt to user profile.

Phase / — Resume Analyzer

Design a production-grade AI Resume Analyzer.

Features:

Resume Upload (PDF/DOCX)

Resume Parsing

Section Detection

ATS Score

Keyword Matching

Formatting Analysis

Grammar Analysis

Missing Skills

Role Match

Company Match

Keyword Suggestions

Bullet Point Improvements

Projects Analysis

Education Analysis

Experience Analysis

Skill Gap

Resume History

Version Comparison

Download Improved Resume

Support targeted company.

Support targeted job role.

Everything should generate actionable suggestions.

Phase 8 — DSA Module

Design a complete DSA Preparation module.

Features:

Company Wise Questions

Topic Wise Questions

Difficulty

Filtering

Search

Bookmarks

Notes

Solved Status

Revision Queue

Random Question

Daily Challenge

Weekly Challenge

Progress

Graphs

Streak

Mock Sheet

Blind 75

NeetCode

Striver Sheet

Solver Sheet

Love Babbar Sheet

Recently Solved

Recommended Next Questions

AI Explanation

Complexity Analysis

Revision Reminder

Everything should scale easily.

Phase 9 — Company Preparation

Create a Company Preparation module.

Each company page should contain:

Overview

Interview Process

Eligibility

CTC

Hiring Pattern

OA Pattern

DSA Questions

Interview Questions

System Design Questions

HR Questions

Behavioral Questions

Core CS Questions

Preparation Timeline

Difficulty Rating

Success Rate

Resources

Company Tags

AI Preparation Plan

Bookmark

Progress

Recommendations

Support 500+ companies.

Phase 10 — Coding Platforms Dashboard

Create an All-in-One Coding Dashboard.

Integrate:

LeetCode

HackerRank

GeeksforGeeks

Codeforces

CodeChef

AtCoder

Display:

Solved Problems

Rank

Contest Rating

Contest Ranking

Contest History

Heatmaps

Progress

Activity Graph

Achievements

Daily Goal

Weekly Goal

Monthly Goal

Personal Best

Support future integrations.

Design API abstraction.

No code.

Phase 11 — AI Personalization

Every recommendation should depend on:

Resume

ATS Score

Target Company

Target Role

Weak Topics

Strong Topics

Solved Questions

Interview Performance

Coding Activity

Generate:

Personalized DSA

Personalized Interview Questions

Weekly Plan

Monthly Plan

Resume Suggestions

Placement Probability

Learning Path

Revision Schedule

Daily Tasks

Everything should update automatically.

Phase 12 — Security

Design production-grade security.

Include:

Authentication

Authorization

RBAC

JWT

Session Management

Input Validation

Rate Limiting

CSRF

SSRF

XSS

SQL Injection

File Upload Security

Audit Logs

API Security

Secrets Management

Vercel Deployment

Supabase Security

Storage Policies

Database Policies

Monitoring

Logging

Backup Strategy

Disaster Recovery

OWASP Top 10 compliance.

Phase 13 — Development Roadmap

Break the implementation into milestones.

Milestone 1

Authentication

Milestone 2

Landing Page

Milestone 3

Dashboard

Milestone 4

Profile

Milestone 5

Resume Analyzer

Milestone 6

DSA Module

Milestone 7

Company Preparation

Milestone 8

Coding Dashboard

Milestone 9

AI Personalization

Milestone 10

Testing

Milestone 11

Optimization

Milestone 12

Deployment on Vercel

For each milestone provide:

Objectives

Tasks

Dependencies

Estimated timeline

Deliverables

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Testing checklist

Definition of Done

Do not generate code until the architecture and planning are complete.

Additional features worth adding

To make your platform stand out from typical placement-prep websites, consider these:

- AI-generated personalized study plans.
- Mock interview sessions with AI feedback.
- Resume version history and comparison.
- Company-specific preparation roadmaps (e.g., Amazon SDE, Microsoft SWE).
- Behavioral interview practice with scoring.
- CS fundamentals tracker (OS, DBMS, CN, OOP).
- Contest calendar and reminders.
- Placement probability trend over time.
- Recruiter-style resume review.
- Gamification with XP, badges, streaks, and leaderboards.
- Exportable progress reports.
- Admin panel for managing questions, companies, and content.
- Offline/PWA support for practicing without a network.

This phased approach keeps the project manageable while producing a scalable application that's well suited for deployment on Vercel and future expansion into a full SaaS platform.

